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Optical window with embedded screen

Abstract

A method of embedding a screen in a substrate includes placing the screen on the substrate, and then melting part of the substrate, so that the screen becomes embedded in the substrate. The melting may involve heating at least part of the screen to melt part of the substrate, or directly heating the part of the substrate. The screen may be a screen of electrically-conductive material, and the heating may be Joule heating in which an electrical current is passed through the screen to heat the screen. Alternatively, the heating may involve microwave, conductive, or laser heating. The produced device of the substrate with an embedded screen may be an optical window with an embedded electromagnetic interference (EMI) screen, may be a touch screen or touch display, or may be a window with an embedded heating element, to give a few non-limiting examples.

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Background/Summary

RELATED APPLICATION (1) This application claims the benefit of U.S. Provisional Application No. 63/334,915, filed Apr. 26, 2022, which is hereby incorporated herein by reference in its entirety.

FIELD
(1) The present disclosure is in the devices having conductive screens.
BACKGROUND
(2) Current methods of attaching electromagnetic interference (EMI) meshes to an optically transparent material can result in partial delamination of the grid or mesh during encapsulation.
SUMMARY
(3) A method embedding a screen or mesh in a substrate includes placing the mesh or screen on a surface of the substrate, and melting a surface portion of the substrate, thereby allowing at least part of the screen to sink into and embed into the substrate.
(4) A device includes a substrate with an embedded mesh or screen.
(5) According to an aspect of the disclosure, a method of embedding a screen in a substrate includes the steps of: placing the screen on the substrate; and melting some of the substrate, resulting in at least part of the screen embedding into the substrate.
(6) According to an embodiment of any paragraph(s) of this summary, the melting includes heating the screen to cause the screen to melt some of the substrate.
(7) According to an embodiment of any paragraph(s) of this summary, the heating the screen includes Joule heating by passing an electrical current through the screen.
(8) According to an embodiment of any paragraph(s) of this summary, the heating the screen includes heating screen and substrate in a heated environment, such as an oven or the like.
(9) According to an embodiment of any paragraph(s) of this summary, the heating the screen includes microwave heating of the screen.
(10) According to an embodiment of any paragraph(s) of this summary, the heating the screen includes laser heating of at least part of the screen.
(11) According to an embodiment of any paragraph(s) of this summary, the laser heating includes selectively varying laser heating of different parts of the screen, thereby varying the embed depth of the different parts of the screen.
(12) According to an embodiment of any paragraph(s) of this summary, the melting includes directly heating the substrate.
(13) According to an embodiment of any paragraph(s) of this summary, the screen is an electrically-conductive material screen.
(14) According to an embodiment of any paragraph(s) of this summary, the electrically-conductive

materials screen is an electrically-conductive mesh formed of electrically-conductive wires.
(15) According to an embodiment of any paragraph(s) of this summary, the heating includes passing an electric current through the electrically-conductive material screen to produce Joule heating.
(16) According to an embodiment of any paragraph(s) of this summary, the electrically-conductive material screen is an electromagnetic interference (EMI) screen.
(17) According to an embodiment of any paragraph(s) of this summary, the electrically-conductive material screen is a metal screen.
(18) According to an embodiment of any paragraph(s) of this summary, the electrically-conductive material screen is a carbon nanotube screen.
(19) According to an embodiment of any paragraph(s) of this summary, the carbon nanotube screen includes carbon nanotube threads or yarns.
(20) According to an embodiment of any paragraph(s) of this summary, the electrically-conductive material screen includes both metal and carbon nanotubes.
(21) According to an embodiment of any paragraph(s) of this summary, the screen is woven.
(22) According to an embodiment of any paragraph(s) of this summary, the screen is nonwoven.
(23) According to an embodiment of any paragraph(s) of this summary, the substrate includes an optical window material.
(24) According to an embodiment of any paragraph(s) of this summary, the substrate includes one or more of the following: zinc sulfide, sapphire, nanocomposite optical ceramic (NCOC), aluminum oxynitride spinel, and yttrium- and/or lanthanum-aluminate glasses.
(25) According to an embodiment of any paragraph(s) of this summary, the substrate includes a non-electrically-conductive (electrical insulating) material.
(26) According to an embodiment of any paragraph(s) of this summary, the substrate includes a polymer.
(27) According to an embodiment of any paragraph(s) of this summary, the method produces an optical window with an embedded EMI screen.
(28) According to an embodiment of any paragraph(s) of this summary, the method produces a touch screen.
(29) According to an embodiment of any paragraph(s) of this summary, the method produces a touch panel.
(30) According to an embodiment of any paragraph(s) of this summary, the method produces a window with a surface heater.
(31) According to an embodiment of any paragraph(s) of this summary, the method produces a window with a defroster or defogger.
(32) According to an embodiment of any paragraph(s) of this summary, the heating the screen includes heating in an inert atmosphere that does not chemically react with the conductive material during the heating.
(33) According to an embodiment of any paragraph(s) of this summary, the heating the screen includes heating in an inert gas atmosphere.
(34) According to an embodiment of any paragraph(s) of this summary, the heating the screen includes heating in an atmosphere that contains argon.
(35) According to an embodiment of any paragraph(s) of this summary, the heating the screen includes heating in an atmosphere that is essentially or substantially free of oxygen.
(36) According to an embodiment of any paragraph(s) of this summary, the heating the screen includes heating in a nitrogen atmosphere.
(37) According to an embodiment of any paragraph(s) of this summary, a method further includes depositing electrically-conductive material along edges of a surface of the substrate.
(38) According to an embodiment of any paragraph(s) of this summary, the depositing the electrically-conductive material includes depositing metal along the edges.

(39) According to an embodiment of any paragraph(s) of this summary, the depositing the electrically-conductive material includes depositing the electrically-conductive material along separate edges on opposite sides of the surface of the substrate.
(40) According to an embodiment of any paragraph(s) of this summary, the depositing occurs after masking the surface to leave the edges exposed.
(41) According to an embodiment of any paragraph(s) of this summary, the depositing occurs before the placing the screen on the substrate.
(42) According to an embodiment of any paragraph(s) of this summary, the placing includes placing the screen in contact with the edges.
(43) According to an embodiment of any paragraph(s) of this summary, the placing includes draping the screen over the substrate.
(44) According to an embodiment of any paragraph(s) of this summary, the placing includes placing with end portions of the screen overhanding and extending beyond edges of the substrate.
(45) According to an embodiment of any paragraph(s) of this summary, a method further includes cutting/trimming the end portions of the screen.
(46) According to an embodiment of any paragraph(s) of this summary, the cutting/trimming of the end portions occurs after the heating of the screen.
(47) According to an embodiment of any paragraph(s) of this summary, a method further includes masking edges of a surface of the substrate by placing masking material over the edges, after the placing the screen on the substrate.
(48) According to an embodiment of any paragraph(s) of this summary, the masking material clamps the screen to the substrate.
(49) According to an embodiment of any paragraph(s) of this summary, the masking material clamps the screen to the electrically-conductive material along edges.
(50) According to an embodiment of any paragraph(s) of this summary, the masking material is an electrically-conductive masking material.
(51) According to an embodiment of any paragraph(s) of this summary, the heating includes applying an electrical current through electrically-conductive masking material and the screen to produce Joule heating in the screen.
(52) According to an embodiment of any paragraph(s) of this summary, the masking material is a metal material.
(53) According to another aspect of the disclosure, a device produced using the method of any paragraph(s) of this summary.
(54) According to yet another aspect of the disclosure, a device includes: a substrate; and a screen at least partially embedded in the substrate.
(55) According to an embodiment of any paragraph(s) of this summary, the device further includes electrically-conductive material along edges of a surface of the substrate.
(56) According to an embodiment of any paragraph(s) of this summary, the electrically-conductive material includes metal along the edges.
(57) According to an embodiment of any paragraph(s) of this summary, the electrically-conductive material is along separate edges on opposite sides of the surface of the substrate.
(58) According to an embodiment of any paragraph(s) of this summary, the device further includes electrically-conductive masking material covering at least in part the electrically conductive material along the edges.
(59) According to an embodiment of any paragraph(s) of this summary, the screen extends between the electrically-conductive material along the edges, and the electrically-conductive masking material.
(60) According to an embodiment of any paragraph(s) of this summary, the device is an optical window with an embedded EMI screen.
(61) According to an embodiment of any paragraph(s) of this summary, the device is a touch

screen.
(62) According to an embodiment of any paragraph(s) of this summary, the device is a touch panel.
(63) While a number of features are described herein with respect to embodiments of the disclosure; features described with respect to a given embodiment also may be employed in connection with other embodiments. The following description and the annexed drawings set forth certain illustrative embodiments of the disclosure. These embodiments are indicative, however, of but a few of the various ways in which the principles of the disclosure may be employed. Other objects, advantages, and novel features according to aspects of the disclosure will become apparent from the following detailed description when considered in conjunction with the drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The annexed drawings, which are not necessarily to scale, show various aspects of the disclosure.
(2) FIG. **1** is a high-level flow chart of a method according to an embodiment.
(3) FIG. **2** is an oblique schematic view illustrating a step of the method of FIG. **1**.
(4) FIG. **3** is an oblique schematic view illustrating another step of the method of FIG. **1**.
(5) FIG. **4** is an oblique schematic view illustrating another step of the method of FIG. **1**.
(6) FIG. **5** is an oblique schematic view illustrating another step of the method of FIG. **1**.
(7) FIG. **6** is a side schematic view illustrating another step of the method of FIG. **1**.
(8) FIG. **7** is a side schematic view illustrating another step of the method of FIG. **1**.
(9) FIG. **8** is a side schematic view illustrating an alternate heating step of the method of FIG. **1**.
(10) FIG. **9** is an oblique schematic view illustrating a device produced according to an embodiment.
DETAILED DESCRIPTION
(11) A method of embedding a screen in a substrate includes placing the screen on the substrate, and then melting part of the substrate, so that the screen becomes embedded in the substrate. The melting may involve heating at least part of the screen to melt part of the substrate, or directly heating the part of the substrate. The screen may be a screen of electrically-conductive material, and the heating may be Joule heating in which an electrical current is passed through the screen to heat the screen. Alternatively, the heating may involve microwave, conductive, or laser heating. The produced device of the substrate with an embedded screen may be an optical window with an embedded electromagnetic interference (EMI) screen, may be a touch screen or touch display, or may be a window with an embedded heating element, to give a few non-limiting examples.
(12) FIG. **1** shows steps of a method **10** for embedding a screen in a substrate. FIGS. **2-8** illustrate various steps (and alternatives) of the method **10**. In step **12**, illustrated in FIG. **2**, electrically-conductive material edge (or peripheral) portions **42** and **44** are deposited along opposite edges **46** and **48** on a surface **50** of a substrate **40**.
(13) The substrate **40** may be any of a variety of suitable substrates. For example the substrate **40** may be a material suitable of an optical window, transparent to light of a desired range of visible or nonvisible wavelengths, for example for use in connection with sensor, such as an infrared (IR) sensor, that may be part of a flight vehicle. Examples of suitable optical window materials may include zinc sulfide (ZnS), sapphire, nanocomposite optical ceramic (NCOC), aluminum oxynitride spinel, and yttrium- and/or lanthanum-aluminate glasses. Such optical window materials may have suitable coatings on them, for example antireflection coatings such as dual antireflection coatings, and/or coatings to resist abrasion and/or other wear, such as rain (and/or sand) enhanced protective coatings. Such coatings may involve a stack of layers of different materials, for example a stack where different materials alternate, one atop of another.

(14) The substrate **40** may be other types of materials as well, such as any of a variety of suitable glasses, such as for windows for ordinary uses. As another alternative, the substrate **40** may be a suitable polymer material. Broadly, the substrate **40** may be a suitable electrically-nonconductive (electrically-insulating) material.

(15) The edge portions **42** and **44** may be made of any of a variety of suitable materials, for example suitable electrically-conductive metals such as copper. The portions **42** and **44** may be deposited by any of a variety of suitable deposition processes, such as electrochemical deposition, soldering, sputtering, or physical deposition. Masking may be involved in the depositing of the edge portions **42** and **44**.

(16) The edge portions **42** and **44** are illustrated as along the parallel opposite edges **46** and **48** of the surface **50**. More broadly, the portions may be along parts or the whole of a perimeter of a surface of the substrate **40**, which may have any of a variety of shapes.

(17) Referring back to FIG. **1**, in step **14** of the method **10** a screen **52** may be transferred onto the substrate surface **50**, as illustrated in FIGS. **3** and **4**. The screen **52** may be draped onto the surface **50**, with ends **56** and **58** of the screen **52** overhanging the substrate **40**, and extending beyond the edges **46** and **48**.

(18) The screen **52** may have any of a variety of suitable characteristics, and may be made of any of a variety of suitable materials. The screen **52** may be made of woven or nonwoven material. The screen **52** may be an electrically-conductive material screen. For example the screen **52** may be a mesh of electrically-conductive wires, for example electrically-conductive metal wires forming a metal screen. Alternatively the screen **52** may be a carbon nanotube (CNT) screen, for example including carbon nanotube threads or yarns. As another alternative the screen **52** may include a combination of metal and CNTs, for example a combination of conductive metal particles and CNTs.

(19) In step **16** (FIG. **1**), masking material **62** and **64** is applied over the edge portions **42** and **44**, as illustrated in FIG. **5**. The masking material **62** and **64** protects the underlying portions **42** and **44** during subsequent processing steps. The masking material **62** and **64** clamps the screen **52** against the edge portions **42** and **44**, holding the screen **52** in place. The masking material **62** and **64** may be electrically-conductive material, for example acting as an electrical contact or busbar for passing current to the screen **52** and to the edge portions **42** and **44**. The masking material **62** and **64** may be made of any suitable metal or other conductive material. Mechanical fixturing or taping may be used for putting the masking material **62** and **64** in place, and/or for fixing the screen **52** in place.

(20) In step **18** (FIG. **1**) a surface portion of the substrate **40** is melted, which allows at least a portion of the screen **52** to sink into and embed in the substrate **40**, below the substrate surface **50**. One embodiment of this process is illustrated in FIGS. **6** and **7**, in which an electrical potential from an electrical power source **70**. This passes an electrical current through the screen **52** via the masking material **62** and **64**. This causes resistive heating (Joule heating) in the screen **52** that melts/softens a top portion **72** of the substrate **40** that is adjacent to the screen **52**. This allows at least part of the screen **52** to sink into and embed in the top substrate portion **72**.

(21) The heating of step **18** (FIG. **1**) may be performed in an inert atmosphere, an atmosphere that does not chemically react with the electrically-conductive material of the screen **52** during the heating. The inert atmosphere may include an inert gas, for example argon. More broadly the atmosphere may be substantially free of oxygen or other gaseous materials that would chemically react with the material of the screen **52** during the heating. For example the heating may be performed in a nitrogen atmosphere.

(22) The resulting embedded screen **52** is shown in FIG. **7**, with the middle portion of the screen **52**, between the edge portions **42** and **44**, embedded in the substrate top portion **72** below the substrate surface **50**. As shown in FIG. **7**, the overhanging ends **56** and **58** (FIG. **6**) of the screen **52** may be trimmed, such as in step **20** (FIG. **1**), either before or after the melting and embedding process.

(23) FIG. **8** shows an alternative method of the melting of the top substrate portion **72**. heating of the screen **52**, and/or by directly heating the top substrate portion **72**. The heating may include application of energy **80** to the screen **52** and/or the substrate top surface **50**. The energy **80** may include any of a variety of suitable forms of energy, such as microwaves, conduction, and/or lasers. The energy **80** may indirectly heat the substrate portion **72** by first heating the screen **52**, which in turns heats (and melts) the top substrate portion **72**. Alternatively or in addition the energy may directly heat the substrate portion **72**. The energy **80** may be applied uniformly to the top of the substrate **40**, or may be applied nonuniformly, such as only on selected parts of the upper surface, and/or with different intensities and/or durations for different parts of the upper surface. This nonuniformity may be achieved by masking and/or by other control, such as by applying energy only in narrow beams that are smaller than the top surface. The nonuniform heating may achieve effects where only parts of the screen **52** are embedded, and/or where different parts of the screen **52** are embedded to different depths within the substrate **40**.
(24) FIG. **9** shows an example of a device **100** produced according to various embodiments of the method **10** (FIG. **1**) described above and illustrated in FIGS. **2-8**. It should be appreciated that variations in the method, for example reordering steps as appropriate, and/or omitting unnecessary steps. The produced device **100** having a substrate **140** with an embedded screen **152** may be an optical window with an embedded electromagnetic interference (EMI) screen, may be a touch screen or touch display, or may be a window with an embedded heating element, to give a few non-limiting examples. Electrically-conductive material portions **162** and **164** may be retained on the device to make electrical connection with the screen **152**.
(25) Although the disclosure has been shown and described with respect to a certain embodiment or embodiments, equivalent alterations and modifications will occur to others skilled in the art upon the reading and understanding of this specification and the annexed drawings. In particular regard to the various functions performed by the above described elements (components, assemblies, devices, compositions, etc.), the terms (including a reference to a “means”) used to describe such elements are intended to correspond, unless otherwise indicated, to any element which performs the specified function of the described element (i.e., that is functionally equivalent), even though not structurally equivalent to the disclosed structure which performs the function in the herein illustrated exemplary embodiment or embodiments of the disclosure. In addition, while a particular feature of the disclosure may have been described above with respect to only one or more of several illustrated embodiments, such feature may be combined with one or more other features of the other embodiments, as may be desired and advantageous for any given or particular application.

Claims

1. A method of embedding a screen in a substrate, the method comprising: depositing electrically-conductive material along edges of a surface of the substrate; placing the screen on the substrate such that the screen contacts the electrically-conductive material along the edges of the surface of the substrate; and melting a portion of the substrate, resulting in at least part of the screen embedding into the substrate, wherein melting the portion of the substrate includes microwave heating of the screen.
2. The method of claim 1, wherein the heating the screen includes heating the screen and substrate in a heated environment.
3. The method of claim 1, wherein melting the portion of the substrate includes directly heating the substrate.
4. The method of claim 1, wherein the substrate includes a non-electrically-conductive material.
5. The method of claim 1, wherein depositing the electrically-conductive material occurs after masking the surface to leave the edges exposed.